

KENDRIYA VIDYALAYA SANGATHAN, PATNA REGION**CLASS X****FIRST PRE-BOARD 2024-25****SUBJECT: ARTIFICIAL INTELLIGENCE (417)****(QUESTION PAPER)****TIME: 2 HOURS****MAX MARKS:50****General Instructions:**

- 1. Please read the instructions carefully.**
- 2. This Question Paper consists of 21 questions in two sections: Section A & Section B.**
- 3. Section A has Objective type questions whereas Section B contains Subjective type questions.**
- 4. Out of the given (5 + 16 =) 21 questions, a candidate has to answer (5 + 10 =) 15 questions in the allotted (maximum) time of 2 hours.**
- 5. All questions of a particular section must be attempted in the correct order.**
- 6. SECTION A - OBJECTIVE TYPE QUESTIONS (24 MARKS):**
 - i. This section has 05 questions.**
 - ii. Marks allotted are mentioned against each question/part.**
 - iii. There is no negative marking.**
 - iv. Do as per the instructions given.**
- 7. SECTION B – SUBJECTIVE TYPE QUESTIONS (26 MARKS):**
 - i. This section has 16 questions.**
 - ii. A candidate has to do 10 questions.**
 - iii. Do as per the instructions given.**
 - iv. Marks allotted are mentioned against each question/part.**

SECTION A: OBJECTIVE TYPE QUESTIONS

Q1	Answer any 4 out of the given 6 questions on Employability Skills (1 x 4 = 4 marks)	
(i)	Which of these is a positive (good) facial expression? A) Frowning while concentrating B) Maintaining eye contact C) Smiling continuously D) Rolling up your eyes	1

(ii)	Stress is identified as a condition in which a person is..... A) Tensed B) Worried C) Both of these D) None of these	1
(iii)	What does the 'S' in SMART goals stand for? A) Smart B) Strategy C) Stupid D) Specific	1
(iv)	Choose the strong password out of the following: A) 12345 B) Aprfd@2024 C) hello123 D) password	1
(v)	Amul (White revolution in India) is an example of which of these business enterprises (a) Sole proprietorship (b) Partnership (c) Cooperative society (d) Joint stock company	1
(vi)	"Efforts are made to increase the solar power generation so that our electricity needs are met and at the same time we do not pollute the environment or use up natural resources". Which SDG can you relate this statement to? (a) Life on land (b) Clean water and sanitation (c) Affordable and clean energy (d) Reduced inequalities	1
Q2	Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)	
(i)	_____ are made with the intent of accomplishing tasks which either too tedious for humans or which are time-consuming. (A) Machines (B) Gadgets (C) Equipment (D) Tools	1

(ii)	_____ refers to the ability to perceive or infer information, and to retain it as knowledge to be applied toward adaptive behaviors within an environment or context. (A) Behavior (B) Nature (C) Habits (D) Intelligence	1
(iii)	Ms. Rani is intelligent in performing calculations and logical reasoning. This ability refers to (A) Linguistic Intelligence (B) Spatial Visual Intelligence (C) Mathematical Intelligence (D) Existential Intelligence	1
(iv)	Speech recognition, Image recognition, and the ability to take action are examples of (A) to perceive, understand, and act (B) reasoning and planning (C) learning and adaption (D) none of these	1
(v)	_____ includes our past experience, intuition, knowledge, and self-awareness (A) habits (B) behaviour (C) information (D) nature	1
(vi)	When a machine possesses the ability to mimic human traits like making decisions, predicting the future, learning, and improving on its own said to have (A) Informative Intelligence (B) Knowledge Intelligence (C) Artificial Intelligence (D) Emotional Intelligence	1
3.	Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)	
(i)	-----help show proportions and percentages between categories, by dividing a circle into proportional segments. (A) Pie Charts (B) Bar Graph (C) Line Chart	1

	(D) Scatter Chart	
(ii)	Which of the following is not a component of project cycle? (A) Problem Scoping (B) Data Acquisition (C) Evaluation (D) Clustering	1
(iii)	-----is a process of understanding the reliability of any AI model, based on outputs by feeding the test dataset into the model and comparing it with actual answers. (A) Evaluation (B) Regression (C) Supervised Learning (D) Unsupervised Learning	1
(iv)	Data science is the process of diverse set of data through? A. Organizing data B. Processing data C. Analysing data D. All of the above	1
(v)	Which of the following step is performed by the data scientist after acquiring the data? A. Data Replication B. Data Integration C. Data Cleansing D. All of the Mentioned	1
(vi)	Which of the following is good way of performing experiments in data science? A. Measure variability B. Generalize to the problem C. Have Replication D. All of the above	1
4	Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)	
(i)	RGB Images are coloured images made up of three primary colours (A) Red, Green and Blue. (B) Red, Green and Brown. (C) Red, Green and Black. (D) None of these	1

(ii)	A pixel value generally is a byte image ie stored as an 8 bit integer with range starting from 0-255. (A) 0 is for Black Colour and 255 is for white colour (B) 0 is for White Colour and 255 is for white Black (C) 0 is for Blue Colour and 255 is for red colour (D) 0 is for red Colour and 255 is for blue colour	1
(iii)	-----application finds its use in smartphones, social media, School attendance systems etc (A) Facial Recognition (B) Medical Imaging (C) Face Filters (D) Google's Search by Image	1
(iv)	What is a chatbot called which uses simple FAQs without any intelligence? (A) Smart Chatbot (B) Script Chatbot (C) AI Chatbot (D) ML Chatbot	1
(v)	What is the process of extracting emotions within a text data using NLP called? (A) Sentiment Analysis (B) Emotional Data Science (C) Emotional Processing (D) Emotional Classification	1
(vi)	What is NLTK tool in Python? (A) Natural Linguistics Tool (B) Natural Language Toolkit (C) Neutral Language Kit (D) Neutral Language Toolkit	1
5	Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)	
(i)	What do we call the process of dividing a string into component words ? (A) Regression (B) Word Tokenization (C) Classification (D) Clustering	1

(ii)	What is the stem of the word “Making” ? (A) Mak (B) Make (C) Making (D) Maker	1
(iii)	While evaluating a model`s performance, recall parameter considers_____ i. False positive ii. True Positive iii. False negative iv. True negative Choose the correct option: A. only (i) B. (ii) and (iii) C. (iii) and (iv) D. (i) and (iv)	1
(iv)	Amit was learning the conditions that make up the confusion matrix. He came across a scenario in which the machine that was supposed to predict an animal was always predicting not an animal. What is this condition called? (A) False Positive (B) True Positive (C) False Negative (D) True Negative	1
(v)	Which of the following talks about how true the predictions are by any model? (A.) Accuracy (B.) Reliability (C.) Recall (D.)F1 Score	1
(vi)	_____ is the process of understanding the reliability of any AI model based on output by feeding the test dataset. (A) Data Reliability (B) Data feed (C) Model Evaluation (D) None of these	1

SECTION B: SUBJECTIVE TYPE QUESTIONS		
Answer any 3 out of the given 5 questions on Employability Skills (2 x 3 = 6 marks) Answer each question in 20 – 30 words.		
Q.6	What are the different types of communication?	2
Q.7	Write different types of domains of self management skill.	2
Q.8	Differentiate between file and folder.	2
Q.9	Give some advantages of entrepreneurship as a career.	2
Q.10	What is the meaning of sustainable development?	2
Answer any 4 out of the given 6 questions in 20 – 30 words each (2 x 4 = 8 marks)		
Q.11	Mention four examples of artificially intelligent applications in our smartphones.	2
Q.12	Define Visualization in Data Science.	2
Q.13	What is Data science? Give an example of it.	2
Q.14	Define computer Vision and mention any two applications of computer vision	2
Q.15	What are some of the applications of chatbots in health care ?	2
Q.16	<i>We must keep in mind that it is not advisable to use the data that we used to create the model to evaluate it. Why?</i>	2
Answer any 3 out of the given 5 questions in 50– 80 words each (4 x 3 = 12 marks)		
Q.17	How intelligent robots are helping us in accomplishing dangerous jobs?	4
Q.18	Differentiate between Supervised Learning and unsupervised Learning.	4
Q.19	Elaborate evaluation in brief.	4
Q.20	What is the difference between how humans interpret communication and how NLP interpret?	4
Q.21	Evaluation is basically done by two things. Explain in details.	4